

# **SIMATS ENGINEERING**

## **ASSESSMENT TOOL 2**

**COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN**  
**COURSE CODE: CSA1153**

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### **COURSE OUTCOME COVERED**

**CO5:** Analyze object-oriented software using quality assurance and usability testing Techniques (BL4, BL5)  
Assessment Tool Weightage: 25%

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**QUESTIONS: 05**

**Total Marks:25**

Annexure A

### **Error Analysis Task**

#### **Q3. Scenario: Inventory Management System**

Modules:

Add Item, Update Stock, Remove Item

#### **Question:**

Write pseudocode to generate and execute **test cases** for stock update functionality, including validation of item availability, quantity limits, and system response. (BL3)

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#### **Code of Conduct:**

*I Y. Abhay –(192411219) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

#### **Signature of the student:**

**Y. Abhay**

## RUBRICS FOR CALCULATION:

### Error Analysis Task

| Criteria                   | Weightage | Level 4 – Exemplary                                                 | Level 3 – Proficient                                 | Level 2 – Developing                                  | Level 1 – Beginning                           |
|----------------------------|-----------|---------------------------------------------------------------------|------------------------------------------------------|-------------------------------------------------------|-----------------------------------------------|
| Error Identification       | 20%       | Accurately identifies all errors with clear understanding of issues | Identifies most errors with minor omissions          | Identifies limited errors; some are missed            | Fails to identify key errors                  |
| Root Cause Analysis        | 30%       | Clearly explains underlying causes with strong technical reasoning  | Explains causes with moderate clarity                | Partial explanation; weak reasoning                   | Unable to identify causes of errors           |
| Correction & Justification | 25%       | Provides correct solutions with strong justification and logic      | Provides correct corrections with some justification | Corrections are partially correct or weakly justified | Incorrect or no corrections provided          |
| Application of Concepts    | 15%       | Effectively applies relevant concepts to analyze and resolve errors | Applies concepts with minor errors                   | Limited application; requires guidance                | Unable to apply concepts                      |
| Clarity & Presentation     | 10%       | Presents analysis clearly, logically, and systematically            | Generally clear with minor issues                    | Lacks clarity or proper structure                     | Poor presentation and difficult to understand |

## Q3. Inventory Management System

### Modules:

- Add Item
- Update Stock
- Remove Item

### Pseudocode to Generate and Execute Test Cases for Stock Update Functionality

#### Algorithm

START

Create Test Cases

TC1: Valid Item and Valid Quantity

TC2: Item Not Available

TC3: Quantity Exceeds Maximum Limit

TC4: Negative Quantity

TC5: Zero Quantity

FOR each Test Case DO

    INPUT ItemID

    INPUT UpdateQuantity

    IF ItemID does not exist THEN

        DISPLAY "Item Not Available"

        Test Case = PASS

    ELSE

        IF UpdateQuantity <= 0 THEN

            DISPLAY "Invalid Quantity"

            Test Case = PASS

        ELSE IF UpdateQuantity > MAX\_LIMIT THEN

            DISPLAY "Quantity Exceeds Limit"

            Test Case = PASS

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ELSE
    CurrentStock = GetStock(ItemID)
    NewStock = CurrentStock + UpdateQuantity

    Update Database with NewStock

    DISPLAY "Stock Updated Successfully"

    IF Database Updated THEN
        Test Case = PASS
    ELSE
        Test Case = FAIL
    ENDIF

ENDIF

ENDIF

ENDIF

END FOR

Generate Test Report

STOP
```

## Test Case Table

| Test Case ID | Input                         | Expected Result            |
|--------------|-------------------------------|----------------------------|
| TC1          | Item Exists, Quantity = 20    | Stock Updated Successfully |
| TC2          | Invalid Item ID               | Item Not Available         |
| TC3          | Quantity = 5000 (Above Limit) | Quantity Exceeds Limit     |
| TC4          | Quantity = -10                | Invalid Quantity           |
| TC5          | Quantity = 0                  | Invalid Quantity           |

## Example Execution

### Test Case 1

Input:

ItemID = 101

Quantity = 20

Output:

Stock Updated Successfully

Status: PASS

### **Test Case 2**

Input:

ItemID = 999

Quantity = 10

Output:

Item Not Available

Status: PASS

### **Test Case 3**

Input:

ItemID = 101

Quantity = 5000

Output:

Quantity Exceeds Limit

Status: PASS

### **Test Case 4**

Input:

ItemID = 101

Quantity = -10

Output:

Invalid Quantity

Status: PASS

## **Flow of Testing**

Start

|

Generate Test Case

|

Check Item Availability

|

+---- No ----> Item Not Available

|

Yes

|

Check Quantity

|

+---- Invalid ----> Error Message

|

Valid

|

Update Stock

|

Verify Database Update

|

Generate Result

|

Stop

## Conclusion

The above pseudocode generates and executes test cases for the **Stock Update** module by validating **item availability**, **quantity limits**, and **system responses**. This ensures accurate inventory updates, prevents invalid entries, and improves the reliability and quality of the Inventory Management System.